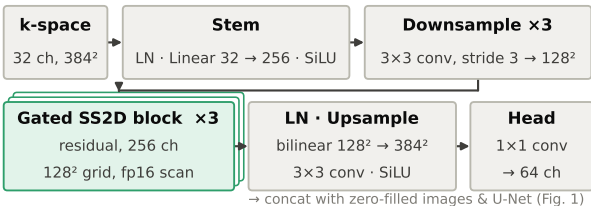
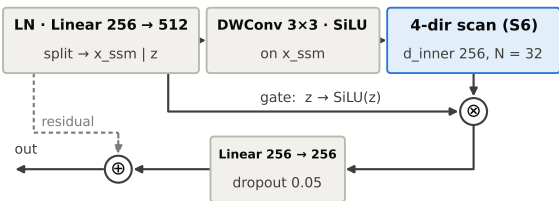


Enhanced SS2D — drop-in replacement for f_θ

arm 34.2M parameters, of which SSM stack 3.1M · 2.84 h/epoch, 80 epochs



Inside one block (256 $\rightarrow$ 256 ch)



vs. controlled SS2D (Fig. 2b): gating $y \odot \text{SiLU}(z)$ restored · 3 residual blocks
· bottleneck lifted (20 $\rightarrow$ 64 ch, d_{inner} 128 $\rightarrow$ 256, d_{state} 16 $\rightarrow$ 32)
· coarse 128^2 scan keeps the epoch time (2.84 h vs. 3.07 h)